Assignment 1 - Git & GitHub Fundamentals Report Template

- Full Name:
- Student ID: 162 542
- Course Code & Section: CSC 3324 01
- GitHub Username: mimou777

1. Repository Setup

```
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER
$ git init
Initialized empty Git repository in C:/Users/hp/Desktop/STARTER/.git/

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        app.js
        app.log
        assets/
        debug.log
        important.log
        index.html
        local.env
        notes.txt
        style.css

nothing added to commit but untracked files present (use "git add" to track)
```

```
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add README.md .gitignore

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add index.html
git commit -m "Add main HTML structure"
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it
[main (root-commit) db8bcbd] Add main HTML structure
 3 files changed, 34 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html
```

Here are the screen shot that show the command i used to initiliaze

2. README & Commit History

README.md M

README.md > # My Git Session 1 Project > ## Example Code

```
# My Git Session 1 Project

## Student Information
- Name: kadmi mimi
- GitHub Username: mimou777

## Description
This project is created to practice Git and GitHub fundamentals.

## Technologies
- HTML
- CSS
- JavaScript
- Git
- GitHub

## Example Code

```javascript
console.log("Hello Git!");
```

GitHub Profile

[Visit my GitHub profile](https://github.com/Amramix108)

Project Image

![Project Banner](assets/project-banner.svg)
```

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```
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[main (root-commit) db8bcbd] Add main HTML structure
 3 files changed, 34 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 index.html

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add style.css
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git touches it
```

3. Selective Staging

```
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add style.css
warning: in the working copy of 'style.css', LF will be replaced by CRLF the next time Git touches it

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add README.md

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git add README.md

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git commit -m "Update README with learning progress"
[main 7189f65] Update README with learning progress
 2 files changed, 44 insertions(+)
 create mode 100644 style.css

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git switch -c feature/about
Switched to a new branch 'feature/about'

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (feature/about)
$ git add README.md
```

4. Branch & Merge

B

```
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git switch -c feature/about
Switched to a new branch 'feature/about'

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (feature/about)
$ git add README.md
git commit -m "Add About section to README"
[feature/about 7cc02a2] Add About section to README
 1 file changed, 1 insertion(+), 1 deletion(-)

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (feature/about)
$ git switch main
Switched to branch 'main'

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git merge feature/about
Updating 7189f65..7cc02a2
Fast-forward
 README.md | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git branch
  feature/about
* main
```

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5. Remote Changes

```
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git remote set-url origin https://github.com/mimou777/git-session1-your-github-mimou777.git

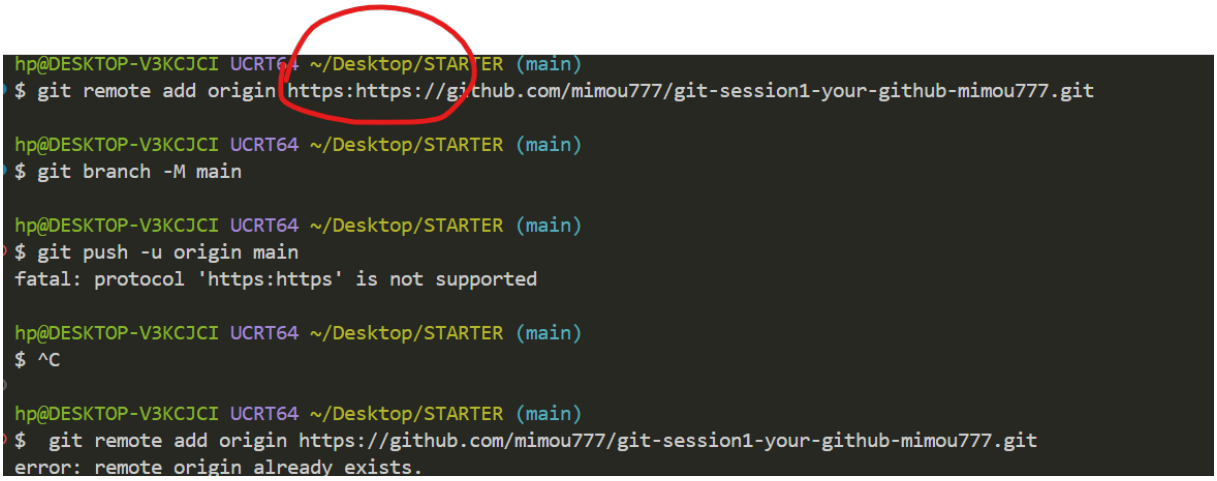
hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git remote -v
origin  https://github.com/mimou777/git-session1-your-github-mimou777.git (fetch)
origin  https://github.com/mimou777/git-session1-your-github-mimou777.git (push)

hp@DESKTOP-V3KCJCI UCRT64 ~/Desktop/STARTER (main)
$ git push -u origin main
info: please complete authentication in your browser...
```

In Git, commit saves your changes locally on your computer, while push uploads those committed changes to the remote repository on GitHub. Fetch downloads information about new changes from GitHub without integrating them into your current working branch, while pull downloads and integrates those changes into your local project.

6. Fair Use of AI

AI did help me solve somme mistakes i did like the one below and it helpt me better understand how git works :



7. Sources & Reflection

Useful sources

- Git Official Documentation
- GitHub Docs – Getting Started with Git
- Atlassian – Git Tutorials

Hardest concept

The hardest concept to understand was the difference between `git fetch` and `git pull`, because both download updates from a remote repository, but only pull integrates those changes into the current branch. If Git behaves unexpectedly, I would first check the repository status using `git status` and verify which branch I am currently working on using `git branch`.

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